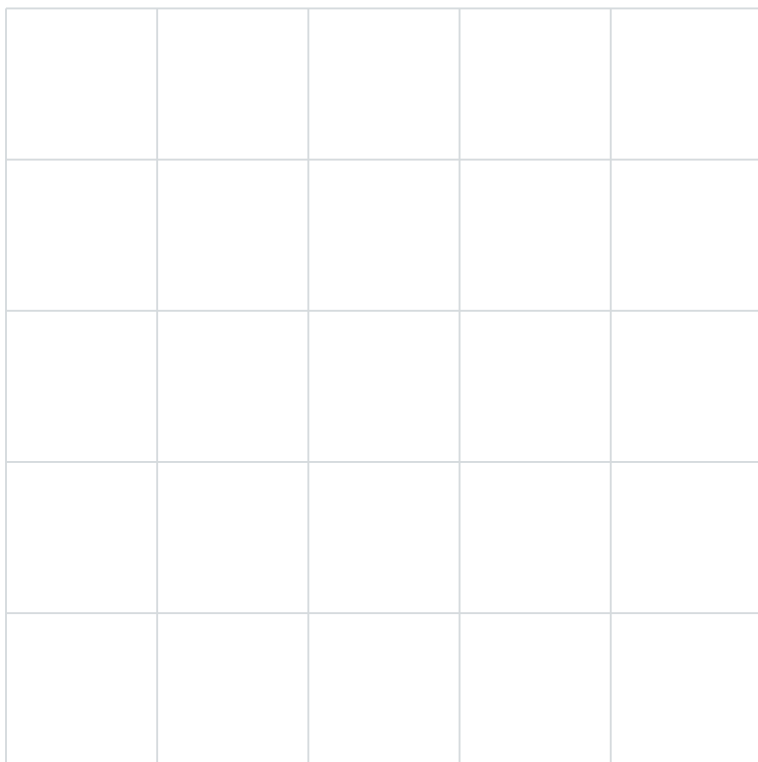


Problem 1: Build every rectangle with whole-number sides at least 1 and $A + B = 6$. Find its area in square tiles. What are the smallest and largest possible areas? Someone uses A and B each from 1 to 5 and promises an area from 1 to 25. Can either end happen with $A + B = 6$? Make the tightest honest range.



One square is one tile.

A	B	Area

All cards in one row describe the SAME unchanged hidden length. An end may lie anywhere in a range, including its edges and between ticks.

Problem 2: In the first row all cards are true: find every possible length. In each other row, exactly one card is false. Which card could it be? Give a possible hidden length for each choice you keep.

2 to 7

4 to 9

5 to 6



1 to 4

3 to 8

5 to 9



1 to 4

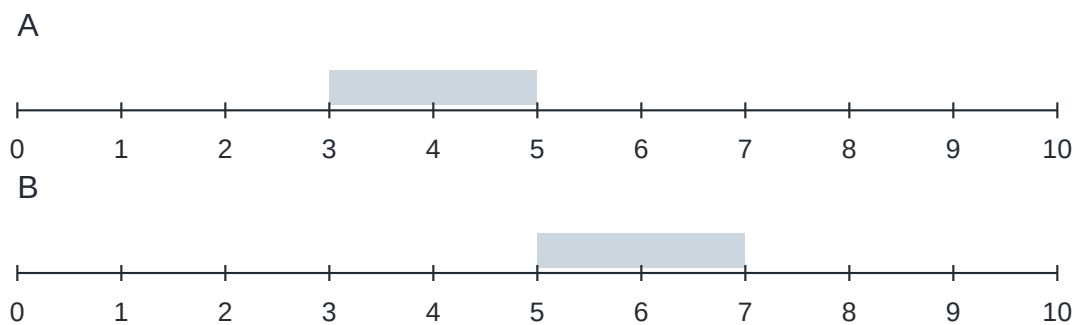
6 to 8

7 to 9



Align both starts at 0. Each length may vary separately anywhere in its band, including the edges. The end gap is the distance between the ends, whichever strip is longer.

Problem 3: For each pair, find the closest and farthest possible ends. Can A be longer, can B be longer, and can they match? Then invent two range cards that always keep the gap at most 2 units, while either strip can be longer.



My cards:
